THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

IRINGA, LINDI AND MOROGORO FORM FOUR REGIONAL MOCK EXAMINATIONS, 2024

CODE: 041

BASIC MATHEMATICS

TIME: 3 HOURS

May, 2024

INSTRUCTIONS

1. This paper consists of sections A and B with a total of **fourteen (14)** questions.
2. Answer **ALL** questions in all sections.
3. Each question in section A carries **six (6)** marks while each question in section B carries **ten (10)** marks.
4. All necessary working and answers for each question must be shown clearly.
5. Mathematical tables and non-programmable calculators may be used.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer booklet(s).

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SECTION A (60 Marks)

Answer **all** questions in this section

1. (a) Find the difference between the smallest and largest number from the given numbers: $0.\dot{3}, \frac{3}{4}, \frac{2}{5}$ and $20\%\ of\ \frac{5}{6}$.

   (b) Without using Mathematical table, evaluate the value of the expression $\frac{0.00075\times3000}{0.00005\times(6.0\times10^4)}$, leaving your answer in standard form.

2. (a) (i) Find the value of x in the equation $3^{x+8} - 30 = 51$.

   (ii) If $p = 2\sqrt{3}$ and $q = -3\sqrt{2}$, find the value of $p^2 + q^2$.

   (b) Solve the equation $3 - n\log 2 = \log 250$.

3. (a) Given P = {multiples of 5} and Q = {even numbers between 10 and 22}. Find all the subsets of a set $P \cap Q$

   (b) In a class of 120 students, $33\frac{1}{3}\%$ learn English, 50% learn Kiswahili and 25% learn both Kiswahili and English. By using Venn diagram, find;

   (i) The number of students who learn English but not Kiswahili.

   (ii) The probability that the student selected randomly from the class learns neither Kiswahili nor English.

4. (a) The line (L_1) perpendicular to the y −axis passes through the point $P(1, 5)$ and intersect with line (L_2) whose equation is $2x - 3y - 7 = 0$ at point Q. Find;

   (i) The coordinate of Q.

   (ii) The equation of a line (L_3) parallel to (L_2) and passes through the midpoint of $\overline{PQ}$.

   (b) Given three vectors $\boldsymbol{a} = 6\boldsymbol{i} - 2\boldsymbol{j}, \boldsymbol{b} = 2\boldsymbol{i} + \boldsymbol{j}$ and $\boldsymbol{c} = 3\boldsymbol{i} - 5\boldsymbol{j}$. Find the magnitude in the direction of the vector $\boldsymbol{d}$ if $\boldsymbol{d} = \frac{1}{2}\boldsymbol{a} + 3\boldsymbol{b} - \boldsymbol{c}$.

5. (a) The wheel of a car has an outer radius of 28 cm. What times the wheel turns for a journey of one Kilometre?

   (b) Given the similar triangle, ΔABC and ΔAXY and $\overline{XY}$ is parallel to $\overline{BC}$. If the area of $\Delta AXY = 4cm^2$, $\overline{AB} = 3cm$ and $\overline{AX} = 2cm$, find the area of ΔABC.

6. (a) The time (T) in seconds that Mary takes to return home from school varies inversely with her average speed (V) in metres per second. If Mary gets back home in half an hour at an average speed of $10m/s$, determine her average speed if she wants to get home in 18 minutes.

   (b) A piece of cloth 37.5 metres long were cut into equal pieces each weighing 75 centimetres long. How many pieces were obtained?

7. (a) The ratio of the number of girls to that of boys in a school is 7:12. If the number of boys in the school is 1380, find the number of girls in the school.

   (b) Miss Ngollo commenced business on 1st February 2024 with cash capital of 600,000 Tanzanian shillings.

February 2:	Bought furniture for cash	50,000
3:	Bought goods for cash	385,000
4:	Paid carriage inwards	 25500
5:	Sold goods for cash	 145,000
7:	Paid rent	 30,000
18:	Cash sales	125,000

20:	Paid wages	50,000
22:	Sold goods on credit to Mr. Singano.....	120,000
25:	Paid for advertisement	20,000
28:	Mr. Singano repaid a loan cash.	100,000
29:	Withdraw for personal use	75,000

Prepare and close the cash account for the above transactions.

8. (a) A conference hall with capacity of 456 seats has 27 seats in the first row and 2 seats more in each preceding row. Determine the total rows the conference hall has.

(b) With reason state whether the series $1-\frac{1}{3}+\frac{1}{9}-\ldots\ldots\ldots\ldots-\frac{1}{243}$ is an arithmetic progression or a geometric progression, and hence find the number of its terms.

9. (a) (i) A tree which is 19cm high, casts the shadow with length 8cm long. Find the distance between the end of the shadow to the top of the tree.

(ii) Evaluate the trigonometric expression $\frac{\cos 120^0 \sin 150^0 + \tan 45^0}{\sin 300^0}$ without using mathematical table. Leave your answer in a simplest surd form.

(b) A boat at point C is observed from two towns A and B which are 3 Km apart. If $\angle CAB = 60^0$ and $\angle CBA = 75^0$, how far is the boat from town B? (Leave your answer correct to two decimal places).

10. (a) The difference of the number increased by 20% and the same number decreased by 15% is 14. Find the number.

(b) A stationery dealer bought a number of notebooks for 3600/=. If each notebook had cost 50/= less he could have bought six more notebooks for the same amount of money. How many notebooks did he buy? (using the quadratic formula).

SECTION B (40 Marks)

Answer **all** questions

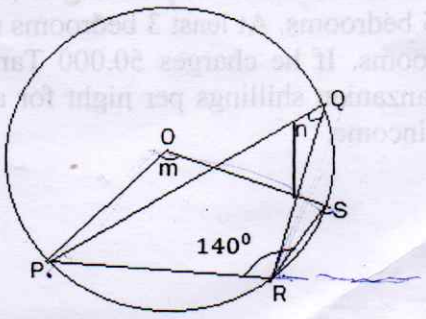
11. (a) In a Mathematics test, the following were scored:

48	47	42	67	73	50	76	47	44	57	45	58	44	54
58	56	66	45	67	32	43	71	48	64	52	42	54	62
49	34	35	46	86	37	47	54	45	60	64	44		

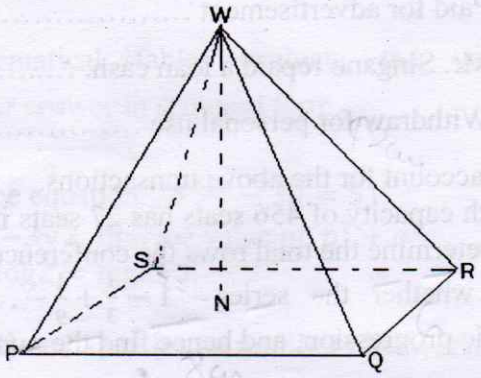
(i) Construct a frequency distribution table with eight (8) class width starting with the smallest score.

(ii) Draw the cumulative frequency curve and from it estimate the median for the distribution.

(b) In the following figure, $\widehat{PRS} = 140^0$. Find the size of the angles m and n.



12. (a) The following figure represents a right pyramid $PQRSW$ whose base $PQRS$ has length of 8 cm and the width of 6 cm and each slant edge is 12 cm. If the vertex W is vertically above at the centre N,



From the figure above;

(i) Name the angle formed between the line $\overline{SW}$ and the base.

(ii) Find the size of the angle obtained in part (a) (i) above.

(iii) Calculate the volume of the pyramid.

(b)Tabora is located at $(5^0S,\ 32^0E)$ on the surface of the Earth. Calculate the radius of the parallel of latitude passes through Tabora. Use 6370 Km as the radius of the Earth.

13. (a) A farmer has 83000 Tanzanian shillings for buying, planting and growing fertilizers. If he buys 2 bags of planting fertilizers and 2 bags of growing fertilizers, he remains with 9000 Tanzanian shillings. If he buys 1 bag of planting fertilizers and 3 bags of growing fertilizers he spends all the money. Using the inverse matrix method, find the price of each bag of planting fertilizers and each bag of growing fertilizers.

(b) The transformation linear equations for a rotation of a point $P(x, y)$ through an angle 270^0 is given as $\begin{cases} x' = x\cos 270^0 - y\sin 270^0 \\ y' = x\sin 270^0 + y\cos 270^0 \end{cases}$, Find its rotation matrix.

(c) A translation $T(a, b)$ takes the coordinates $(-2, -3)$ onto $(10,\ 3)$, find where it will take the coordinates $(3,\ 5)$.

14. (a) A mobile phone plan from TTCL network costs 500 Tanzanian shillings a month. The plan includes 25 free minutes and charges 60 Tanzanian shillings for each additional minute of usage. The monthly charges are a function of the number of minutes used, given by $C(t) = \begin{cases} 500 \quad if \quad 0 \le t \le 25 \\ 500 + 60(t - 25) \quad if \quad t > 25 \end{cases}$

Determine the cost that the plan charges for;

(i) 48 seconds.

(ii) 30 minutes.

(b) The owner of CREST hotel in Mwanza is planning to build another hotel of an extension containing not more than 15 bedrooms. At least 3 bedrooms must be single rooms and at least 2 rooms must be double rooms. If he charges 50,000 Tanzanian shillings per night for a double room and 30,000 Tanzanian shillings per night for a single room, find the best plan which gives him maximum income.